

Chapter 9 Strings

Section 9.2 The String Class

1. Suppose `s` is a string with the value "java". What will be assigned to `x` if you execute the following code?

```
char x = s.charAt(4);
```

- a. 'a'
- b. 'v'
- c. Nothing will be assigned to `x`, because the execution causes the runtime error `StringIndexOutOfBoundsException`.

2. Which of the following statements is preferred to create a string "Welcome to Java"?

- a. `String s = "Welcome to Java";`
- b. `String s = new String("Welcome to Java");`
- c. `String s; s = "Welcome to Java";`
- d. `String s; s = new String("Welcome to Java");`

3. What is the output of the following code?

```
public class Test
{
    public static void main(String[] args)
    {
        String s1 = "Welcome to Java!";
        String s2 = s1;

        if (s1 == s2)
            System.out.println("s1 and s2 reference to the same String object");
        else
            System.out.println("s1 and s2 reference to different String objects");
    }
}
```

- a. s1 and s2 reference to the same String object
- b. s1 and s2 reference to different String objects

4. What is the output of the following code?

```
public class Test
{

    public static void main(String[] args)
    {
        String s1 = "Welcome to Java!";
        String s2 = "Welcome to Java!";

        if (s1 == s2)
            System.out.println("s1 and s2 reference to the same String object");
        else
            System.out.println("s1 and s2 reference to different String objects");
    }
}
```

- a. s1 and s2 reference to the same String object
- b. s1 and s2 reference to different String objects

5. What is the output of the following code?

```
public class Test
{
    public static void main(String[] args)
```

```

{
    String s1 = new String("Welcome to Java!");
    String s2 = new String("Welcome to Java!");

    if (s1 == s2)
        System.out.println("s1 and s2 reference to the same String object");
    else
        System.out.println("s1 and s2 reference to different String objects");
}

```

- a. s1 and s2 reference to the same String object
- b. s1 and s2 reference to different String objects

6. Suppose s1 and s2 are two strings. What is the result of the following code?

```

s1.equals(s2) == s2.equals(s1)
a. true
b. false

```

7. What is the output of the following code?

```

public class Test
{
    public static void main(String[] args)
    {
        String s1 = new String("Welcome to Java!");
        String s2 = new String("Welcome to Java!");

        if (s1.equals(s2))
            System.out.println("s1 and s2 have the same contents");
        else
            System.out.println("s1 and s2 have different contents");
    }
}

```

- a. s1 and s2 have the same contents
- b. s1 and s2 have different contents

8. What is the output of the following code?

```

public class Test
{
    public static void main(String[] args)
    {
        String s1 = new String("Welcome to Java!");
        String s2 = s1.toUpperCase();

        if (s1 == s2)
            System.out.println("s1 and s2 reference to the same String object");
        else if (s1.equals(s2))
            System.out.println("s1 and s2 have the same contents");
        else
            System.out.println("s1 and s2 have different contents");
    }
}

```

- a. s1 and s2 reference to the same String object
- b. s1 and s2 have the same contents
- c. s1 and s2 have different contents

9. What is the output of the following code?

```

public class Test
{
    public static void main(String[] args)
    {
        String s1 = new String("Welcome to Java");
        String s2 = s1;

        s1 += "and Welcome to HTML";

        if (s1 == s2)
            System.out.println("s1 and s2 reference to the same String object");
        else
            System.out.println("s1 and s2 reference to different String objects");
    }
}

```

- a. s1 and s2 reference to the same String object
- b. s1 and s2 reference to different String objects

10. Suppose s1 and s2 are two strings. Which of the following statements or expressions are incorrect?

- a. String s = new String("new string");
- b. String s3 = s1 + s2
- c. s1 >= s2
- d. int i = s1.length
- e. s1.charAt(0) = '5'

11. Suppose s1 and s2 are two strings. Which of the following statements or expressions is incorrect?

- a. String s3 = s1 - s2;
- b. boolean b = s1.compareTo(s2);
- c. char c = s1[0];
- d. char c = s1.charAt(s1.length());

12. "abc".compareTo("aba") returns _____.

- a. 1
- b. 2
- c. -1
- d. -2
- e. 0

13. "AbA".compareToIgnoreCase("abC") returns _____.

- a. 1
- b. 2
- c. -1
- d. -2
- e. 0

14. _____ returns true.

- a. "peter".compareToIgnoreCase("Peter")
- b. "peter".compareToIgnoreCase("peter")
- c. "peter".equalsIgnoreCase("Peter")
- d. "peter".equalsIgnoreCase("peter")
- e. "peter".equals("peter")

15. What is the output of the following code?

```

String s = "University";
s.replace("i", "ABC");
System.out.println(s);

```

- a. UnABCversity

- b. UnABCversABCTy
- c. UniversABCTy
- d. University

16. What is the return value of "SELECT".substring(0, 5)?

- a. "SELECT"
- b. "SELEC"
- c. "SELE"
- d. "ELECT"

17. What is the return value of "SELECT".substring(4, 4)?

- a. an empty string
- b. C
- c. T
- d. E

18. Analyze the following code.

```
class Test
{
    public static void main(String[] args)
    {
        String s;
        System.out.println("s is " + s);
    }
}
```

- a. The program has a compilation error because s is not initialized, but it is referenced in the println statement.
- b. The program has a runtime error because s is not initialized, but it is referenced in the println statement.
- c. The program has a runtime error because s is null in the println statement.
- d. The program compiles and runs fine.

19. To check if a string s contains the prefix "Java", you may write

- a. if (s.startsWith("Java")) ...
- b. if (s.indexOf("Java") == 0) ...
- c. if (s.substring(0, 4).equals("Java")) ...
- d. if (s.charAt(0) == 'J' && s.charAt(1) == 'a' && s.charAt(2) == 'v' && s.charAt(3) == 'a') ...

20. To check if a string s contains the suffix "Java", you may write

- a. if (s.endsWith("Java")) ...
- b. if (s.lastIndexOf("Java") >= 0) ...
- c. if (s.substring(s.length() - 4).equals("Java")) ...
- d. if (s.substring(s.length() - 5).equals("Java")) ...
- e. if (s.charAt(s.length() - 4) == 'J' && s.charAt(s.length() - 3) == 'a' && s.charAt(s.length() - 2) == 'v' && s.charAt(s.length() - 1) == 'a') ...

21. Which of the following is the correct statement to return JAVA?

- a. toUpperCase("Java")
- b. "Java".toUpperCase("Java")
- c. "Java".toUpperCase()
- d. String.toUpperCase("Java")

22. Which of the following is the correct statement to return a string from an array a of characters?

- a. toString(a)

- b. new String(a)
- c. convertToString(a)
- d. String.toString(a)

23. Assume s is " abc ", the method _____ returns a new string "abc".

- a. s.trim(s)
- b. trim(s)
- c. String.trim(s)
- d. s.trim()

24. Assume s is "ABCABC", the method _____ returns a new string "aBCaBC".

- a. s.toLowerCase(s)
- b. s.toLowerCase()
- c. s.replace('A', 'a')
- d. s.replace('a', 'A')
- e. s.replace("ABCABC", "aBCaBC")

25. Assume s is "ABCABC", the method _____ returns an array of characters.

- a. toChars(s)
- b. s.toCharArray()
- c. String.toChars()
- d. String.toCharArray()
- e. s.toChars()

26. _____ returns a string.

- a. String.valueOf(123)
- b. String.valueOf(12.53)
- c. String.valueOf(false)
- d. String.valueOf(new char[]{'a', 'b', 'c'})

27. The following program displays _____.

```
public class Test
{
    public static void main(String[] args)
    {
        String s = "Java";
        StringBuilder buffer = new StringBuilder(s);
        change(s);
        System.out.println(s);
    }
    private static void change(String s)
    {
        s = s + " and HTML";
    }
}
```

- a. Java
- b. Java and HTML
- c. and HTML
- d. nothing is displayed

28. What is displayed by the following statement?

```
System.out.println("Java is neat".replaceAll("is", "AAA"));
```

- a. JavaAAANEat
- b. JavaAAA neat
- c. Java AAA neat
- d. Java AAANEat

29. What is displayed by the following code?

```
public static void main(String[] args)
```

```

{
    String[] tokens = "Welcome to Java".split("o");
    for (int i = 0; i < tokens.length; i++)
    {
        System.out.print(tokens[i] + " ");
    }
}

```

- a. Welcome to Java
- b. Welc me to Java
- c. Welc me t Java
- d. Welcome t Java

29. What is displayed by the following code?

```

System.out.print("Hi, ABC, good".matches("ABC ") + " ");
System.out.println("Hi, ABC, good".matches(".*ABC.*"));

```

- a. false fasle
- b. true fasle
- c. true true
- d. false true

29. What is displayed by the following code?

```

System.out.print("A,B;C".replaceAll(";", "#") + " ");
System.out.println("A,B;C".replaceAll("[,;]", "#"));

```

- a. A B C A#B#C
- b. A#B#C A#B#C
- c. A,B;C A#B#C
- d. A B C A B C

29. What is displayed by the following code?

```

String[] tokens = "A,B;C;D".split("[,;]");
for (int i = 0; i < tokens.length; i++)
    System.out.print(tokens[i] + " ");

```

- a. A,B;C;D
- b. A B C D
- c. A B C;D
- d. A B;C;D

Section 9.5 The Character Class

30. Which of following is not a correct method in Character?

- a. isLetterOrDigit(char)
- b. isLetter(char)
- c. isDigit()
- d. toLowerCase(char)
- e. toUpperCase()

31. Suppose Character x = new Character('a'), _____ returns true.

- a. x.equals(new Character('a'))
- b. x.compareToIgnoreCase('A')
- c. x.equalsIgnoreCase('A')
- d. x.equals('a')
- e. x.equals("a")

Section 9.6 The StringBuilder/StringBuffer Class

32. Analyze the following code.

```

class Test
{
    public static void main(String[] args)
    {
        StringBuilder strBuf = new StringBuilder(4);
    }
}

```

```
    strBuf.append("ABCDE");  
    System.out.println("What's strBuf.charAt(5)? " + strBuf.charAt(5));  
}
```

```
}
```

- a. The program has a compilation error because you cannot specify initial capacity in the StringBuilder constructor.
- b. The program has a runtime error because because the buffer's capacity is 4, but five characters "ABCDE" are appended into the buffer.
- c. The program has a runtime error because the length of the string in the buffer is 5 after "ABCDE" is appended into the buffer. Therefore, strBuf.charAt(5) is out of range.
- d. The program compiles and runs fine.

33. Which of the following is true?

- a. You can add characters into a string buffer.
- b. You can delete characters into a string buffer.
- c. You can reverse the characters in a string buffer.
- d. The capacity of a string buffer can be automatically adjusted.

34. _____ returns the last character in a StringBuilder variable named strBuf?

- a. strBuf.charAt(strBuf.length() - 1)
- b. strBuf.charAt(strBuf.capacity() - 1)
- c. StringBuilder.charAt(strBuf.length() - 1)
- d. StringBuilder.charAt(strBuf.capacity() - 1)

35. Assume StringBuilder strBuf is "ABCDEFGH", after invoking _____, strBuf contains "AEFG".

- a. strBuf.delete(0, 3)
- b. strBuf.delete(1, 3)
- c. strBuf.delete(1, 4)
- d. strBuf.delete(2, 4)

36. Assume StringBuilder strBuf is "ABCDEFGH", after invoking _____, strBuf contains "ABCRRRRDEFG".

- a. strBuf.insert(1, "RRRR")
- b. strBuf.insert(2, "RRRR")
- c. strBuf.insert(3, "RRRR")
- d. strBuf.insert(4, "RRRR")

37. Assume StringBuilder strBuf is "ABCCEFC", after invoking _____, strBuf contains "ABTTEFT".

- a. strBuf.replace('C', 'T')
- b. strBuf.replace("C", "T")
- c. strBuf.replace("CC", "TT")
- d. strBuf.replace('C', "TT")
- e. strBuf.replace(2, 7, "TTEFT")

38. The StringBuilder methods _____ not only change the contents of a string buffer, but also returns a reference to the string buffer.

- a. delete
- b. append
- c. insert
- d. reverse
- e. replace

39. The following program displays _____.

```
public class Test  
{  
    public static void main(String[] args)
```

```

{
    String s = "Java";
    StringBuilder buffer = new StringBuilder(s);
    change(buffer);
    System.out.println(buffer);
}
private static void change(StringBuilder buffer)
{
    buffer.append(" and HTML");
}
}

```

- a. Java
- b. Java and HTML
- c. and HTML
- d. nothing is displayed

Section 9.7 Command-Line Arguments

40. How can you get the word "abc" in the main method from the following call?

```
java Test "+" 3 "abc" 2
```

- a. args[0]
- b. args[1]
- c. args[2]
- d. args[3]

41. Given the following program:

```

public class Test
{
    public static void main(String[] args)
    {
        for (int i = 0; i < args.length; i++)
        {
            System.out.print(args[i] + " ");
        }
    }
}

```

What is the output, if you run the program using

```
java Test 1 2 3
```

- a. 3
- b. 1
- c. 1 2 3
- d. 1 2

42. Which code fragment would correctly identify the number of arguments passed via the command line to a Java application, excluding the name of the class that is being invoked?

- a. int count = args.length;
- b. int count = args.length - 1;
- c. int count = 0; while (args[count] != null) count ++;
- d. int count=0; while (!(args[count].equals(""))) count ++;

43. Which correctly creates an array of five empty Strings?

- a. String[] a = new String [5];
- b. String[] a = {"", "", "", "", ""};
- c. String[5] a;

d. `String[ ] a = new String [5]; for (int i = 0; i < 5; a[i++] = null);`

44. Identify the problems in the following code.

```
public class Test
{
    public static void main(String argv[])
    {
        System.out.println("argv.length is " + argv.length);
    }
}
```

- a. The program has a compile error because `String argv[]` is wrong and it should be replaced by `String[] args`.
- b. The program has a compile error because `String args[]` is wrong and it should be replaced by `String args[]`.
- c. If you run this program without passing any arguments, the program would have a runtime error because `argv` is null.
- d. If you run this program without passing any arguments, the program would display `argv.length` is 0.

45. Which of the following is the correct header of the main method?

- a. `public static void main(String[] args)`
- b. `public static void main(String args[])`
- c. `public static void main(String[] x)`
- d. `public static void main(String x[])`
- e. `static void main(String[] args)`